

Практическая работа

Задание:

1. Изучите материалы по ссылкам, которые размещены в теме.
2. Отправьте GET и POST запросы на любой веб-ресурс (см. публичные API в материалах темы) в CLI с помощью Telnet или netcat.
3. Отправьте запросы, выполненные по заданию 2, с помощью cURL.
4. Установите Insomnia, Postman, HTTPie Desktop или другой инструмент. Отправьте с его помощью GET-запрос для получения курса одной выбранной валюты за выбранный период (задайте любые значения). Используйте API Банка России: <https://www.cbr.ru/development/sxml/>
5. Запустите два процесса netcat — сервер (netcat -l) и клиент (подключается к localhost). Проверьте работу полученного простейшего чата и продемонстрируйте скриншотами в отчёте.
6. Подготовьте отчёт, который включает описание действий и скриншоты с результатами выполнения.

Результат:

GET запрос:

Выполнен с помощью nc

```
PS C:\Users\stepa> nc -l -p 443
GET /api/users/7 HTTP/1.1
Host: reqres.in

HTTP/1.1 200 OK
Date: Thu, 10 Apr 2025 08:45:38 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 342
Connection: keep-alive
CF-Ray: 92e0f814391c9d3d-DME
Server: cloudflare
Report-To: {"group":"heroku-nel","max_age":3600,"endpoints":[{"url":"https://nel.heroku.com/reports?ts=1743474506&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=cxeZitEcWwUkVs1d1ckPTTRZn4hZB%2Fr42vqP6lsHf0c%3D"}]}
Reporting-Endpoints: heroku-nel=https://nel.heroku.com/reports?ts=1743474506&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=cxeZitEcWwUkVs1d1ckPTTRZn4hZB%2Fr42vqP6lsHf0c%3D
Nel: {"report_to":"heroku-nel","max_age":3600,"success_fraction":0.005,"failure_fraction":0.05,"response_headers":["Via"]}
X-Powered-By: Express
Access-Control-Allow-Origin: *
Etag: W/"156-oMlgkyEs6eb1/fskSNkbJSKBTI8"
Via: 1.1 vegur
Cache-Control: max-age=14400
CF-Cache-Status: REVALIDATED
Accept-Ranges: bytes
server-timing: cfL4;desc="?proto=TCP&rtt=242726&min_rtt=122665&rtt_var=130166&sent=8&recv=7&lost=0&retrans=0&sent_bytes=2822&recv_bytes=706&delivery_rate=33750&cwnd=241&unsent_bytes=0&cid=c24eecdffbf0ceef&ts=4987&x=0"
{"data":{"id":7,"email":"michael.lawson@reqres.in","first_name":"Michael","last_name":"Lawson","avatar":"https://reqres.in/img/faces/7-image.jpg"},"support":{"url":"https://contentcaddy.io?utm_source=reqres&utm_medium=json&utm_campaign=referral","text":"Tired of writing endless social media content? Let Content Caddy generate it for you."}}
```

POST запрос:

```

PS C:\Users\stepa> ncat --ssl reqres.in 443
POST /api/users HTTP/1.1
Host: reqres.in
Content-Length: 36
Content-Type: application/json

{"name": "john", "job": "programmer"}

HTTP/1.1 201 Created
Date: Thu, 10 Apr 2025 08:47:29 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 84
Connection: keep-alive
Cf-Ray: 92e0fac81d1bfc28-AMS
Server: cloudflare
Report-To: {"group":"heroku-nel","max_age":3600,"endpoints":[{"url":"https://nel.heroku.com/reports?ts=1744274849&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=Ibr9GfSkqfKnS0sg1ZfYzMJHOP2FK0m6qgwPU248oRc%3D"}]}
Reporting-Endpoints: heroku-nel=https://nel.heroku.com/reports?ts=1744274849&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=Ibr9GfSkqfKnS0sg1ZfYzMJHOP2FK0m6qgwPU248oRc%3D
Nel: {"report_to":"heroku-nel","max_age":3600,"success_fraction":0.005,"failure_fraction":0.05,"response_headers":["Via"]}
X-Powered-By: Express
Access-Control-Allow-Origin: *
Etag: W/"54-7NM/PcQqIfq2iYG0iZDSjDV5rjo"
Via: 1.1 vegur
Cf-Cache-Status: DYNAMIC
server-timing: cfl4;desc="?proto=TCP&rtt=119251&min_rtt=77634&rtt_var=58839&sent=7&recv=6&lost=0&retrans=0&sent_bytes=2821&recv_bytes=770&delivery_rate=53327&wnd=253&unsent_bytes=0&cid=b4eb6b43f24ca95c&ts=3857&x=0"

{"name":"john","job":"programmer","id":"506","createdAt":"2025-04-10T08:47:29.121Z"}

```

Отправка с помощью cURL

GET запрос:

```

PS C:\Users\stepa> curl reqres.in/api/users/7

StatusCode      : 200
StatusDescription : OK
Content          : {"data":{"id":7,"email":"michael.lawson@reqres.in","first_name":"Michael","last_name":"Lawson","avatar":"https://reqres.in/img/faces/7-image.jpg"},"support":{"url":"https://contentcaddy.io?utm_source=...
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Cf-Ray: 92e0fc699cce2956-AMS
                  Report-To: {"group":"heroku-nel","max_age":3600,"endpoints":[{"url":"https://nel.heroku.com/reports?ts=1744274901&sid=c4c9725f-1a...
Forms           : {}
Headers         : {[Connection, keep-alive], [Cf-Ray, 92e0fc699cce2956-AMS], [Report-To, {"group":"heroku-nel","max_age":3600,"endpoints":[{"url":"https://nel.heroku.com/reports?ts=1744274901&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=zzShmKEWve1EF1509sGGp9qPAELlk1QovJJobxIAKNY%3D"}]}], [Reporting-Endpoints, heroku-nel=https://nel.heroku.com/reports?ts=1744274901&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=zzShmKEWve1EF1509sGGp9qPAELlk1QovJJobxIAKNY%3D]}...]}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 342

PS C:\Users\stepa> |

```

POST запрос:

Сделан через Invoke-WebRequest, т.к. выдавал curl ошибку

```

PS C:\Users\stepa> Invoke-WebRequest -Uri https://reqres.in/api/users -Method POST -ContentType "application/json" -Body
'{"name": "john", "job": "programmer"}'

StatusCode      : 201
StatusDescription : Created
Content         : {"name":"john","job":"programmer","id":"223","createdAt":"2025-04-10T08:55:14.877Z"}
RawContent      : HTTP/1.1 201 Created
                  Connection: keep-alive
                  Cf-Ray: 92e1062c2c679dc7-DME
                  Report-To: {"group":"heroku-nel","max_age":3600,"endpoints":[{"url":"https://nel.heroku.com/reports
?ts=1744275314&sid=c4c972...
Forms           : {}
Headers         : {[Connection, keep-alive], [Cf-Ray, 92e1062c2c679dc7-DME], [Report-To, {"group":"heroku-nel","max_a
ge":3600,"endpoints":[{"url":"https://nel.heroku.com/reports?ts=1744275314&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=ult
BWUbo9QMU%2FgrhqVmaZekyXoXqvV5ehcGwjwLBLhM%3D"}]}]}, [Reporting-Endpoints, her
oku-nel=https://nel.heroku.com/reports?ts=1744275314&sid=c4c9725f-1ab0-44d8-820f-430df2718e11&s=ult
BWUbo9QMU%2FgrhqVmaZekyXoXqvV5ehcGwjwLBLhM%3D]}...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 84

PS C:\Users\stepa> |

```

С помощью Insomnia

GET запрос:

The screenshot shows the Insomnia REST client interface. The top bar indicates a GET request to `https://www.cbr.ru/scripts/XML_dynamic.asp?date_req1=01/01/2023&date_req2=31/12/2023` with a status of 200 OK, 670 ms response time, and 28.9 KB body size. The 'Params' tab is selected, showing query parameters. The 'Preview' tab on the right displays the XML response, which is a currency market dynamic report for the period from 01.01.2023 to 31.12.2023. The XML includes records for various dates in 2023, each containing nominal and unit rates.

Два процесса netcat:

Были открыты два окна в терминале, один был сервером, а другой клиентом.

The screenshot shows a Windows PowerShell terminal window. The user has run the command `ncat -l -p 12345` to start a netcat listener. The terminal output shows the listener waiting for a connection, then receiving a connection from `localhost`. The user then types `Hello!` and the netcat process responds with `Hello!`.